1. ATM Pin Validation

Write a program to simulate an ATM that keeps asking the user for their PIN using a while loop. The loop should terminate once the correct PIN is entered, or after 3 incorrect attempts.

In [47]:
```
# Solution 1
# existing_pin = "1234"
# attempts=0
# while True:
#     pin=input("Please enter your secret pin")
#     if pin == existing_pin:
#         print("Access granted")
#         break
#     else:
#         attempts+=1
#         rem=3-attempts
#         if rem>0:
#             print(f"Incorrect pin, you have {rem} remaining attempts!!")
#         else:
#             print("3 wrong attempts, Access denied")
#             break


try:
    pin=1234
    count=0
    while count<3:
        count+=1
        pin_check=int(input("Please enter your secret pin: "))
        if pin_check==pin:
            print("Access Granted")
            break
        else:
            if count<3:
                print(f"Incorrect entry number: {count}. Attempt remaining: {3-(count)}")
            else:
                print("3 consecutive incorrect entries. Account locked")

except:
    print("Invalid PIN format")
```

```
Incorrect entry number: 1. Attempt remaining: 2
```

```
Incorrect entry number: 2. Attempt remaining: 1
Access Granted
```

2. Password Strength Checker

Write a program that takes a password as input and uses a while loop to ensure it meets the following criteria: At least 8 characters long Contains both letters and numbers Keep prompting the user until a valid password is entered.

In [1]:
```
# Solution 2
while True:
    paswd=input("Please enter the password to check the strength: ")
    if len(paswd)>=8 and paswd.isalnum()==True and " " not in paswd and not paswd.isnumeric() and not paswd.isalpha():
        print("Strong Password")
        break
    else:
        print("The password entered is weak, Please try again with 8 characters long, containining both letters and numbers")
        continue
```

```
The password entered is weak, Please try again with 8 characters long, containining both letters and numbers
Strong Password
```

3. Recharge Reminder

A prepaid mobile plan expires after X days. Use a while loop to count down the days, displaying a reminder each day, until the plan expires. When the count reaches 0, display "Plan Expired."

In [11]:
```
# Solution 3
try:
    rem=int(input("Enter the number of days left for the plan to expire = "))
    if rem<0:
            print("Invalid number of days")
    else:
        while rem>=0:
            if rem>0:
                print(f"{rem} day(s) left before plan expires")
            else:
                print("Plan Expired")
            rem-=1

except ValueError:
    print("Wrong data input")
```

```
5 day(s) left before plan expires
4 day(s) left before plan expires
3 day(s) left before plan expires
2 day(s) left before plan expires
1 day(s) left before plan expires
Plan Expired
```

4. Data Entry Validation

Write a program that continuously asks the user to input a valid email address using a while loop. If the input is invalid, display an error message and prompt again.

In [44]:
```
# Solution 4, this simulation asks the user to enter valid emails continuously, so did not use break to exit the loop
# but used iterations for 10 valid emails only, once 10 valid email is achieved, the loop ends
count=0
while count<10:
    email=input("Please enter email for validation:")
    last=["com","in","org"]
    if " " not in email and "@" in email and "." in email:
        em=email.split(".")
        em=em[-1]
        if em in last:
            print("Valid Email")
            count+=1
            continue
        else:
            print("Incorrect email, Please enter a Valid Email Address")

    else:
        print("Incorrect email, Please enter a Valid Email Address")
```

```
Valid Email
Incorrect email, Please enter a Valid Email Address
Valid Email
Incorrect email, Please enter a Valid Email Address
Incorrect email, Please enter a Valid Email Address
Valid Email
Valid Email
Valid Email
Valid Email
Valid Email
Valid Email
Incorrect email, Please enter a Valid Email Address
```

```
Valid Email
Incorrect email, Please enter a Valid Email Address
Valid Email
```

5. Loan Repayment Simulation

Simulate loan repayment using a while loop. Start with a loan amount, subtract the monthly payment each iteration, and display the remaining balance. Exit the loop when the loan is fully repaid.

In [17]:
```
# Solution 5. This simulation allows for variable monthly payments by requesting input inside the loop.
try:
    loan_amount=float(input("Please enter the loan amount: ₹"))
    i=1
    if loan_amount>0:
        while loan_amount>0:
            emi=float(input("Please enter the monthly installment amount you want to pay: ₹"))
            if loan_amount>emi:
                loan_amount-=emi
                print(f"Installment amount received for \"month: {i}\" is ₹{emi}. Remaining loan balance is: ₹{loan_amount}")
            elif loan_amount<=emi:
                # loan_amount-=emi
                print(f"Loan amount repaid, Loan Balance is ₹ 0.00,  Total loan amount repaid in \"{i}\" month(S).")
                break
            i+=1
    else:
        print("Loan amount is negligible")
except ValueError:
    print("Please enter a valid amount")
```

```
Installment amount received for "month: 1" is ₹50.0. Remaining loan balance is: ₹50.0
Installment amount received for "month: 2" is ₹25.0. Remaining loan balance is: ₹25.0
Loan amount repaid, Loan Balance is ₹ 0.00,  Total loan amount repaid in "3" month(S).
```

6. Quiz Game

Create a simple quiz game where the program keeps asking questions (e.g., "What is 5+3?") using a while loop. Exit the loop if the user answers incorrectly or completes 5 questions correctly.

In [109…
```
# Solution 6
questions=["What is 5+3?","What is the capital of India","What comes after number 9?","How many sides does a triangle have?",
           "What is 45/9 ?"]
```

```
solutions=["8","delhi","10","3","5"]
g=dict(zip(questions,solutions))

counter=0
while counter<len(questions):
    question=questions[counter]
    answer=input(f"Question {counter+1} -> {question}: ").lower().strip()
    if g[question]==answer:
        print(f"{answer.title()} is the correct answer.")
        counter+=1
        if counter==5:
            print("All questions answered correctly!")
            break

    else:
        print(f"{answer} is the wrong answer for Question \"{question}\". Please start the quiz again")
        break
```

```
8 is the correct answer.
Delhi is the correct answer.
10 is the correct answer.
5 is the wrong answer for Question "How many sides does a triangle have?". Please start the quiz again
```

7. Temperature Check

A freezer stops working if the temperature exceeds a threshold (e.g., -5°C). Use a while loop to simulate temperature readings every minute, and display a warning if the threshold is breached.

In [45]:
```
# Solution 7. This simulation allows for different temperatures by requesting input inside the while loop.
try:
    while True:
        temperature=int(input("Please enter the freezer temperature reading: "))
        if temperature>=-5:
            print("WARNING! temperature threshold breached, temperature reading: {temperature}°C")
        else:
            print(f"{temperature}°C does not exceed threshold temperature, working fine.")
            continue


except :
    print("Please Enter the Temperature in Numeric form only")
```

```
WARNING! temperature threshold breached, temperature reading: {temperature}°C
```

```
-6°C does not exceed threshold temperature, working fine.
Please Enter the Temperature in Numeric form only
```

8. Parking Lot Availability

Simulate a parking lot with 10 available spaces. Use a while loop to keep accepting car entries and decrement the space count. Stop the loop when no spaces are left.

In [27]:
```
# Solution 8
space_available=10
while space_available>0:
    car_entry=input("Please enter car details")
    space_available-=1
    print(f"Entry successful for car number {10-space_available}, remaining spaces left {space_available}")
    if space_available==0:
        print("No more spaces left to park a vehicle")
```

```
Entry successful for car number 1, remaining spaces left 9
Entry successful for car number 2, remaining spaces left 8
Entry successful for car number 3, remaining spaces left 7
Entry successful for car number 4, remaining spaces left 6
Entry successful for car number 5, remaining spaces left 5
Entry successful for car number 6, remaining spaces left 4
Entry successful for car number 7, remaining spaces left 3
Entry successful for car number 8, remaining spaces left 2
Entry successful for car number 9, remaining spaces left 1
Entry successful for car number 10, remaining spaces left 0
No more spaces left to park a vehicle
```

9. Toll Booth Counter

Write a program for a toll booth that keeps track of the number of vehicles passing through and stops collecting data when a user enters "stop." Display the total number of vehicles.

In [110…
```
#Solution 9
try:
    vehicle_count=0
    while True:
        vehicle=input("Please enter the vehicle category or number, Please enter \"Stop\" to Stop counting").strip().lower()
        if vehicle!="stop":
            vehicle_count+=1
```

```
        else:
            break
except:
    pass

finally:
    if count>0:
        print(f"Total number of vehicles passed is {vehicle_count}")
    else:
        print("No vehicles passed")
```

```
Total number of vehicles passed is 3
```

10. Shopping Budget Tracker

Simulate a shopping experience where the user has a limited budget. Keep allowing the user to purchase items (by entering their price) using a while loop, and exit the loop when their budget is exhausted.

In [48]:

```
# Solution 10
try:
    budget=float(input("Enter you budget: ₹"))
    if budget>0:
        while budget>0:
            goods_price=float(input("Enter the price of the good: ₹"))
            if budget>=goods_price:
                budget-=goods_price
                if budget>0:
                    print(f"Item purchased worth ₹{goods_price}, remaining balance = ₹{budget}")
                else:
                    print(f"Last item purchased worth ₹{goods_price}. Budget exhausted.")
            elif budget<goods_price:
                print("Exceeding budget cannot buy this stuff")
                continue
    else:
        print("You do not have enough budget to go for a shopping")

except:
    print("Wrong input detected")
```

```
Item purchased worth ₹5.0, remaining balance = ₹5.0
Exceeding budget cannot buy this stuff
Item purchased worth ₹4.0, remaining balance = ₹1.0
Last item purchased worth ₹1.0. Budget exhausted.
```